

Late-stage N-Me selective alkenylation and alkynylation of drugs and unactivated tertiary amines using photoredox catalysis

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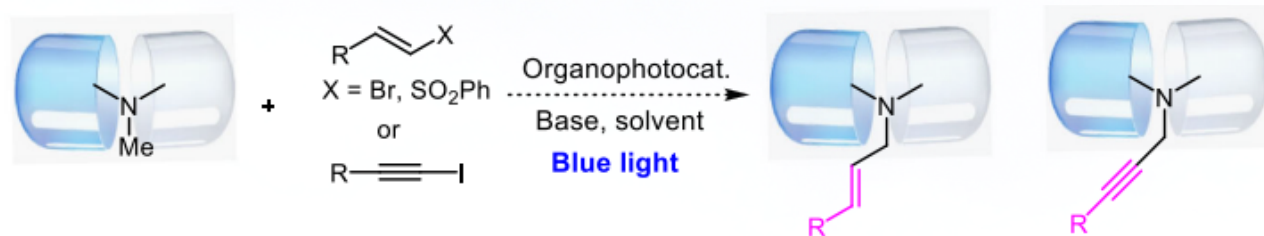
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Introduction

- Amine functionalities are found in >40% of pharmaceutical compounds, and nearly 60% of these amines are tertiary amines.
- Late-stage functionalization allows chemists to modify already developed or marketed drugs.
- Regioselective modification of C–H bonds α to nitrogen is difficult because aliphatic tertiary amines often contain several similar α -amino C–H bonds with similar bond dissociation energies.

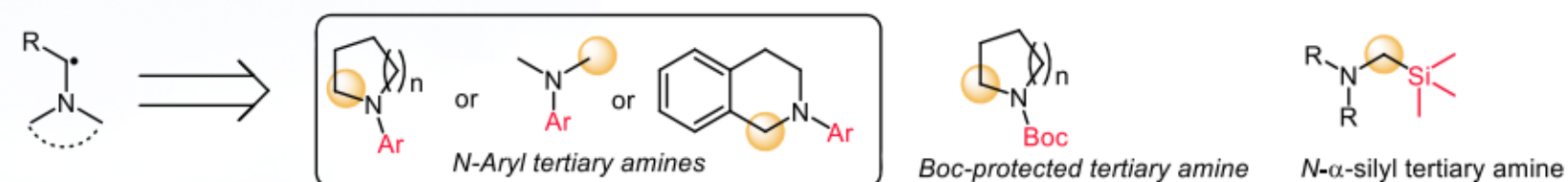
This study's approach

- A visible-light organophotoredox strategy for N-methyl selective alkenylation and alkynylation of aliphatic tertiary amines.
- Introducing vinyl and alkyne groups is useful because it increases structural diversity and enables π – π interactions, bio-orthogonal labelling, click chemistry, and drug conjugation.

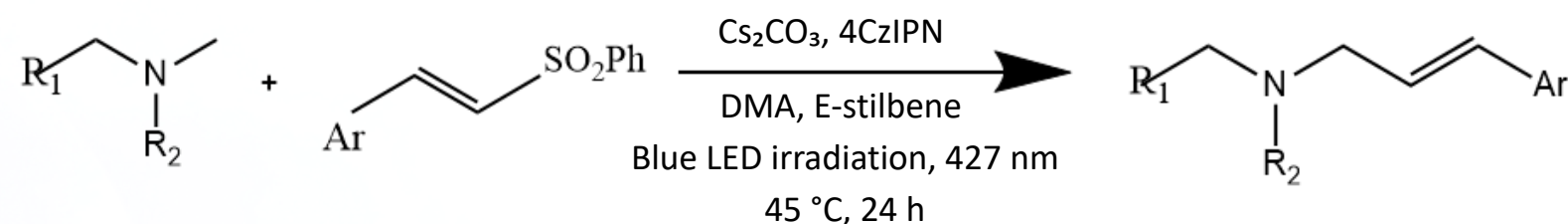


Current limitations and research gap

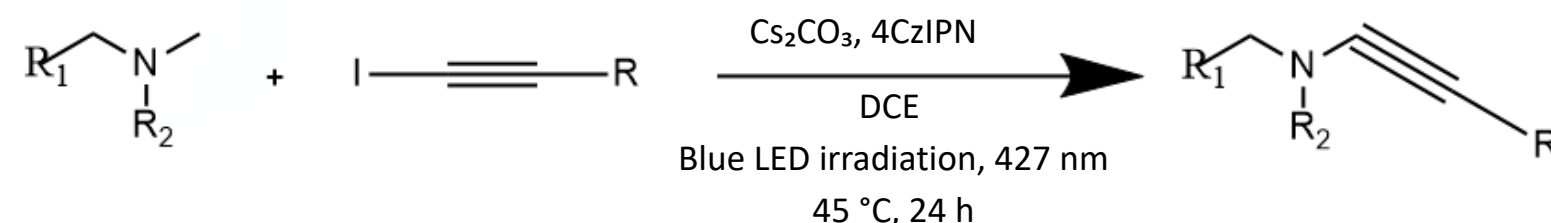
- Many existing methods require N-aryl amines, N-Boc-protected amines, α -silyl amines, transition metals, or pre-activated substrates.
- Transition-metal-free, site-selective alkenylation and alkynylation of unactivated aliphatic tertiary amines and tertiary amine-based marketed drugs remains underexplored.



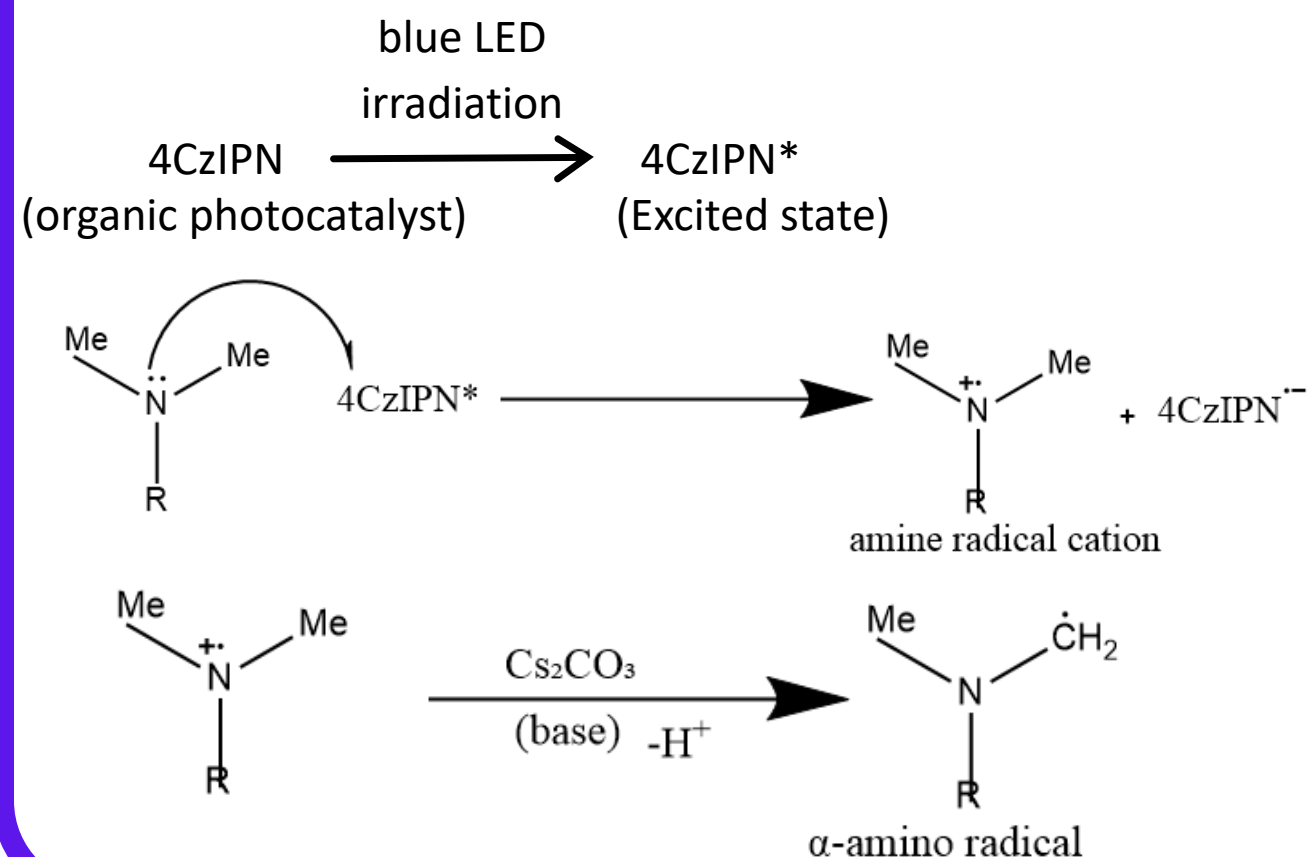
Alkenylation



Alkynylation



Step 1: Radical formation



DFT (Density Functional Theory) /Selectivity

- A proton can theoretically be removed from different α-amino positions, especially the N-Me position or the benzylic position.
- Activation barrier for proton abstraction:

N-Me position: 7.5 kcal/mol,

Benzylic pathway: 10.4 kcal/mol

- N-Me-derived α-amino radical forms more easily and is kinetically favoured.
- Benzylic radical is more thermodynamically stable because it's stabilized by the nearby aromatic ring.

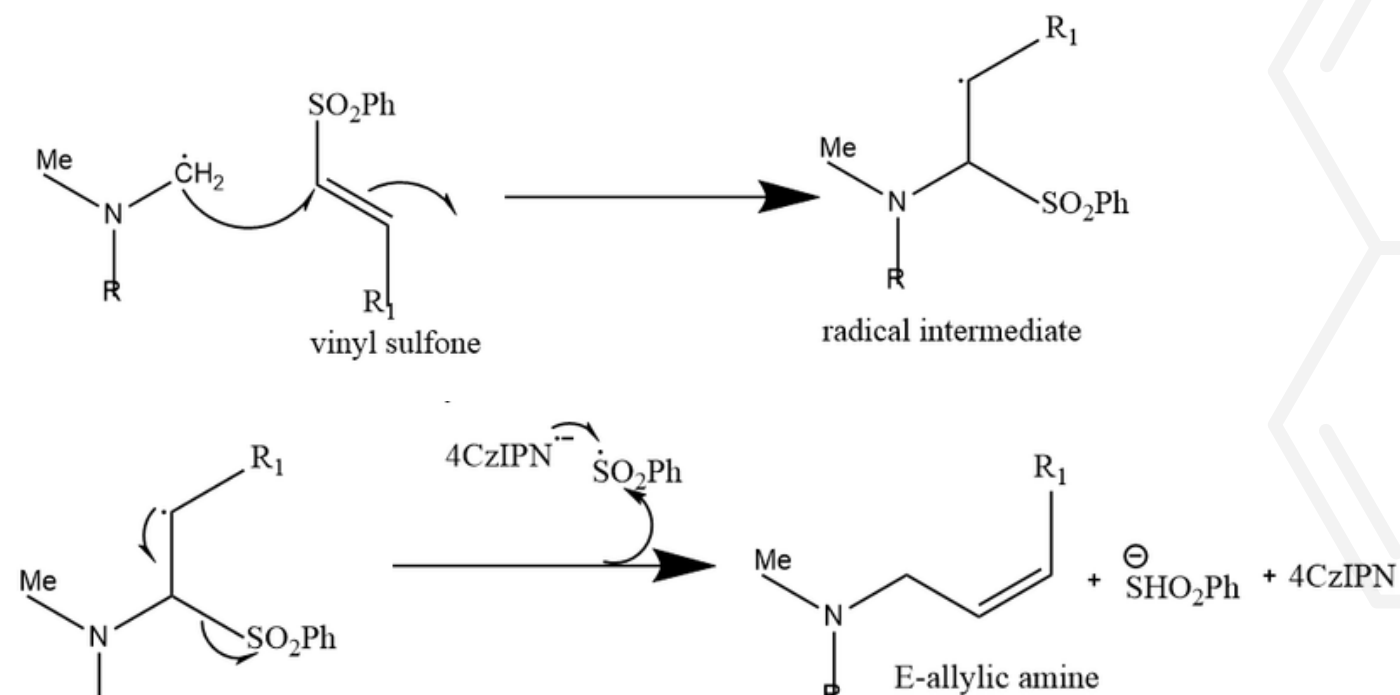
N-Me-derived radical addition to phenyl vinyl sulfone: 7.9 kcal/mol,

Benzylic radical addition pathway: 15.6 kcal/mol.

The observed N-methyl regioselectivity is mainly controlled by the radical addition step.

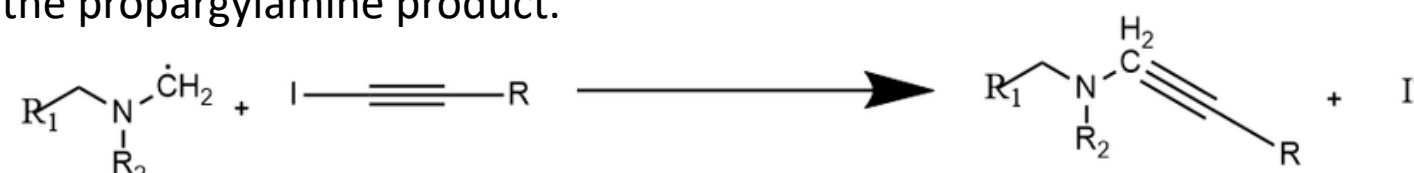
The reaction is controlled mainly by kinetics, not thermodynamic radical stability.

Step 2: Product formation through radical addition



Alkylation

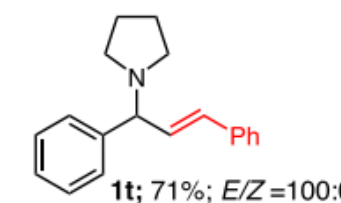
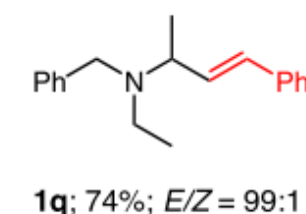
- α-amino radical reacts with an alkynyl iodide, followed by iodide loss, giving the propargylamine product.



Control of E/Z selectivity using E-stilbene

- In the alkenylation reaction, the desired product is the E-allylic amine, but under photoredox conditions the product can undergo unwanted E-to-Z isomerization.
- The likely occurs by an energy transfer process from 4CzIPN* to the alkene product
- .Added E-stilbene as an energy-transfer inhibitor/additive, to suppress this.
- E-stilbene likely competes with the alkenylated product for excited-state energy, undergo its own E-to-Z isomerization, forming Z-stilbene as a by-product.
- This protects the desired E-allylic amine from being excited and isomerized.

1 equivalent of E-stilbene allowed the reaction to achieve almost 100% E-selectivity.



Biological Validation of N-Me Functionalized Imipramine

Analogues

- Imipramine is a marketed tricyclic antidepressant, it is a real drug molecule, that contains a tertiary amine.
- Imipramine is known to increase synaptic serotonin / 5-HT levels by blocking serotonin reuptake.
- This occurs through inhibition of hSERT, the human serotonin transporter, which normally transports serotonin back into presynaptic neurons.
- N-Me selective alkenylation and alkynylation method was used to create new imipramine analogues.
- Analogues were designed to contain extended N-alkyl groups, often with aromatic rings, which could improve binding through extra hydrophobic and π -stacking interactions.

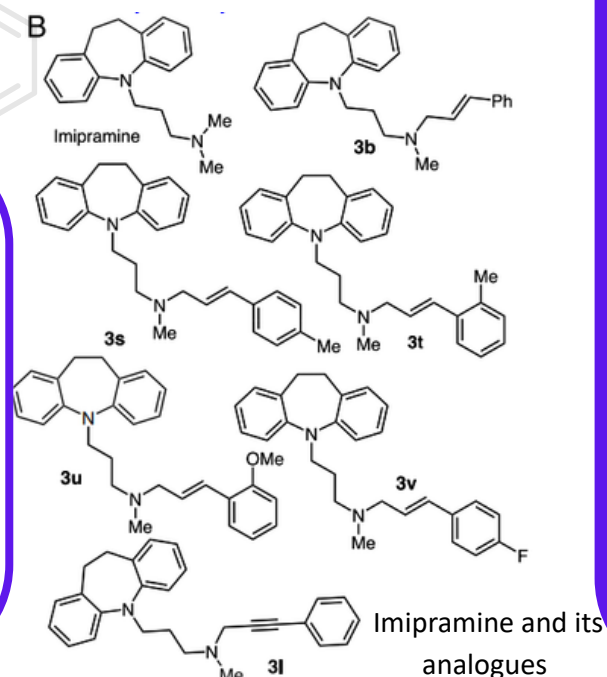
Molecular Docking:

- Predicted whether imipramine analogues could still bind to hSERT.
- Was done using a computer software, not as a wet-lab experiment.
- Designed and modelled 20 imipramine derivatives, based on the known binding mode to hSERT.
- Selected six promising analogues for biological testing, based on docking binding energy.

References

Paul, S., Mahato, S., Mahapatra, S. K., Dey, S., Paul, D., Mondal, S., Naskar, A., Srikar, S., Mohanty, S., Khatun, S., Ghorai, D., Singh, A. K., Das, S., Jana, N. R., Bandyopadhyay, B., Das, A., Roy, L., & Panda, S. (2026b). Late-stage N-Me selective alkenylation and alkynylation of drugs and unactivated tertiary amines using photoredox catalysis. *Nature Communications*, 17(1), 2695-. <https://doi.org/10.1038/s41467-026-69131-7>

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Drosophila depression model

- Tested whether the selected imipramine analogues showed antidepressant-like biological activity.
- Used *Drosophila melanogaster*, the fruit fly, as the biological model.
- The model was based on social isolation-induced depression-like behaviour.
- Newly emerged adult male flies were kept alone in single-housing tubes for 5 days.
- Five-day social isolation causes a depression-like state, which is associated with increased male-male aggression.
- Aggression was used as the quantitative behavioural readout.
- After isolation, aggression was measured using an aggression assay, where the number of aggressive bouts was counted over 20 minutes.
- The drugs were delivered through the fly food during the 5-day isolation period.
- Imipramine and its analogues were first dissolved in DMSO (dimethyl sulfoxide), a common organic solvent used to dissolve drug-like molecules.
- DMSO-only-fed isolated flies were used as the control group.
- The DMSO-only-fed isolated flies showed a significantly increased number of aggressive bouts compared with socially housed flies.
- Treatment with imipramine significantly reduced the number of aggressive bouts.

Results:

- The selected imipramine analogues were then compared with imipramine.
- Analogues 3b and 3s were equally effective as imipramine.
- Analogues 3t, 3u, 3v, and especially 3i showed better antidepressant-like activity than Imipramine.

Future Direction:

Further testing is needed in more advanced in vivo mammalian models, such as mice or rats.

Drosophila provides an early biological screening model, but it cannot fully represent human depression or human drug metabolism.

